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Course: CIS 25
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Malik Mays' Midterm Project - Charm CRM

Introduction

For this course's midterm I am presenting the project I have been working on since the start of the semester, Charm CRM. Charm is a CLI based customer relationship management tool written in C++. It's designed for professional freelancers, creative entrepreneurs, small-but-mighty teams and anyone else in need of a lightweight method of organizing contacts while building high value networks. Meet the executive director of your favorite local nonprofit at a volunteer event? Write down their info in charm! Finally make contact with the head of sales at that spice company you'd like to partner with for a new product? Charm! A blackbook for your MacBook, if you will.

The project repo is structured as follows:

Midterm/

- charmCRM (executable)
- CLIUtilities.cpp
- CLIUtilities.h
- Customer.cpp
- Customer.h
- CustomerRepository.cpp
- CustomerRepository.h
- Interaction.cpp
- Interaction.h
- main.cpp
- output.pdf (this document)
- Transaction.cpp
- Transaction.h
- User.cpp
- User.h
- UserRepository.cpp
- UserRepository.h

This project has been a great exercise in the fundamentals of C++ and object oriented programming, and I encountered many challenges including:

- Figuring out what classes needed to exist in order to adequately represent all necessary entities
- Determining what system responsibilities belong to which classes

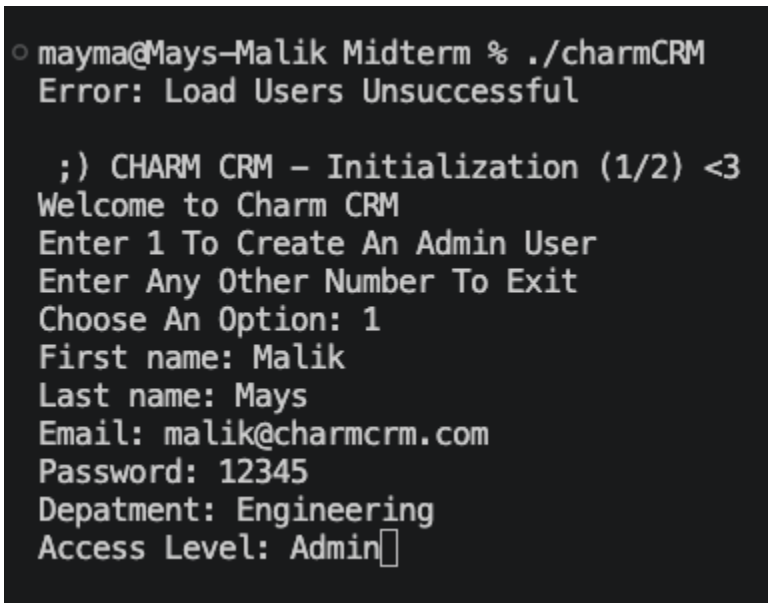
- Passing values from the CLI to constructors appropriately
- Learning when and when not to use const
- Learning the difference between references and pointers and when to use them
- Generating sequential user and customer ID numbers and assigning them procedurally
- Ensuring next_user_id and next_customer_id counters are set to the right number upon loading data
- Customizing the provided Binary Search example for my program

The project adequately demonstrates the following midterm requirements:

- Data types and data sizes (int, bool, etc.)
- File types (.h, .cpp, .csv)
- Use of pointers (and references)
- Arrays (std::vector)
- Binary search (and linear search)
- Use of strings
- File I/O (both read from .csv and write to .csv)
- At least 2 classes (7 classes)

Initialization

On first use, Charm will walk new users through an initialization process that includes the setup of a user with Admin access …



… and another with Standard access

```
 ;) CHARM CRM - Initialization (2/2) <3
Enter 1 To Create A Standard User
Enter Any Other Number To Exit
Choose An Option: 1
First name: Jake
Last name: Thoms
Email: jake@charmcrm.com
Password: 12345
Depatment: Product
Access Level: Standard
```

Note: No system data exists yet so the line “Error: Load Users Unsuccessful” is intended behavior.

Log In

Success moves users to the log in screen where they must provide their email and password to proceed.

```
 ;) CHARM CRM - Login (1/3) <3
Enter 1 To Log In
Enter 2 To Exit
Choose Option: 1

 ;) CHARM CRM - Login (2/3) <3
Enter Email: jake@charmcrm.com

 ;) CHARM CRM - Login (3/3) <3
Enter Password: 12345
```

Standard Menu

Logging in as a Standard user takes them to a restricted menu with read-only options:

- View customer records
- View user records
- Find customers by IID
- Find users by ID
- Log out

```
 ;) CHARM CRM <3
1. View Customer Records
2. View User Records
3. Find Customer By ID
4. Find User By ID
5. Log Out
Choose an option:
```

Admin Menu

Logging in as an Admin brings users to an expanded menu that allows them to:

- Create a new customer record
- Create a new user record
- View customer records
- View user records
- Find customers by ID
- Find users by ID
- Create a new transaction
- Create a new interaction
- Save and log out

```
 ;) CHARM CRM - Login (1/3) <3
Enter 1 To Log In
Enter 2 To Exit
Choose Option: 1

 ;) CHARM CRM - Login (2/3) <3
Enter Email: malik@charmcrm.com

 ;) CHARM CRM - Login (3/3) <3
Enter Password: 12345
```

```
 ;) CHARM CRM <3
1. Create New Customer Record
2. Create New User Record
3. View Customer Records
4. View User Records
5. Find Customer By ID
6. Find User By ID
7. Create New Transaction
8. Create New Interaction
9. Save And Log Out
Choose an option:
```

Create a New Customer Record

Let's create some customers. Enter "1" and create a new customer record:

```
First name: Whitney
Last name: Houston
Organization: Arista Records
Email: whitney@arista.com
Phone number: 5551234567
City: East Orange
State: NJ
Address: 456 Voice ave.
Representative: malik@charmcrm.com
```

And a few more…

```
First name: Dinah
Last name: Washington
Organization: Mercury Records
Email: dinah@mercury.com
Phone number: 5556789999
City: Tuscaloosa
State: AL
Address: 321 Sangin ln.
Representative: malik@charm.com
```

```
First name: Beyonce
Last name: Knowles
Organization: Parkwood Entertainment
Email: bey@parkwoodent.com
Phone number: 5557654321
City: Houston
State: TX
Address: 999 Wow st.
Representative: malik@charm.com
```

Create a New Transaction

Now that we have some customers let's add a transaction. Say Beyonce (customer #3) paid us $500 for a 1yr company license to Charm. Back at the Admin menu, let's enter "7":

```
Enter customer ID: 3
Enter transaction type: Cash
Enter transaction date: 10/30/2026
Enter transaction amount: $500
```

Create a New Interaction

After Beyoncé (customer #3) bought that license she sent an email with mention of someone she knew who would love to use Charm. Let's make a note of that interaction. Back at the Admin menu let's enter "8":

```
Enter customer ID: 3
Enter interaction type: Email
Enter interaction date: 10/30/2026
Enter interaction notes: Beyonce says ask Solange about Charm
```

View Customer Records

Back at the main menu, let's enter "3" to display info on all of our customers so far:

```
Customer ID: 1
Customer Name: Whitney Houston
Customer Organization: Arista Records
Customer Email: whitney@arista.com
Customer Location: 456 Voice ave. East Orange, NJ
Customer Representative: malik@charmcrm.com
Is Active Customer: 1
Liftime Customer Value: $0

Customer ID: 2
Customer Name: Dinah Washington
Customer Organization: Mercury Records
Customer Email: dinah@mercury.com
Customer Location: 321 Sangin ln. Tuscaloosa, AL
Customer Representative: malik@charm.com
Is Active Customer: 1
Liftime Customer Value: $0

Customer ID: 3
Customer Name: Beyonce Knowles
Customer Organization: Parkwood Entertainment
Customer Email: bey@parkwoodent.com
Customer Location: 999 Wow st. Houston, TX
Customer Representative: malik@charm.com
Is Active Customer: 1
Liftime Customer Value: $500
```

Notice that customer #3 has a lifetime customer value of $500 after entering Beyoncé's purchase.

Create a New User Record

Let's say Charm hired a new employee. Back at the Admin menu, let's enter "2" to create a new user of the program:

```
First name: Lily
Last name: Fields
Email: lily@charmcrm.com
Password: 12345
Depatment: Sales
Access Level: Admin
```

View User Records

Let's take a look at info for all of our users, just like we did with customers. At the Admin menu enter "4":

```
User ID: 1
User Name: Malik Mays
User Email: malik@charmcrm.com
User Department: Engineering

User ID: 2
User Name: Jake Thoms
User Email: jake@charmcrm.com
User Department: Product

User ID: 3
User Name: Lily Fields
User Email: lily@charmcrm.com
User Department: Sales
```

Find Customers by ID

If we want info on a specific customer we can search for them using their unique ID number. Because customers are stored sequentially in a vector, the vector can be searched through with binary search executing at O(log n) time. Say we want to search for Dinah Washington, customer #2. At the Admin menu enter "5":

```
Enter Customer ID: 2

Customer ID: 2
Name: Dinah Washington
Email: dinah@mercury.com
Phone: 5556789999
Organization: Mercury Records
Location: 321 Sangin ln. Tuscaloosa, AL
Representative: malik@charm.com
Is Active Customer: 1
Liftime Customer Value: $0
```

Find Users by ID

We can also search for users in similar fashion. Say we want info on Charm's most recent employee, employee #3. At the Admin menu enter "6":

```
Enter User ID: 3

User ID: 3
Name: Lily Fields
Email: lily@charmcrm.com
Department: Sales
```

Saving Data & Exiting Charm

The last option in the Admin menu is "Save And Log Out" which does what it says on the tin. User and customer objects are accessed from the relevant vectors that store them, then their data is accessed and stored in .csv format. At the Admin menu enter "9":

```
Would You Like to Change The Filename [y/n]?
Choose Option: n
Users Saved Successfully
Would You Like to Change The Filename [y/n]?
Choose Option: n
Customer Data Saved Successfully
Thanks for using CHARM!
```

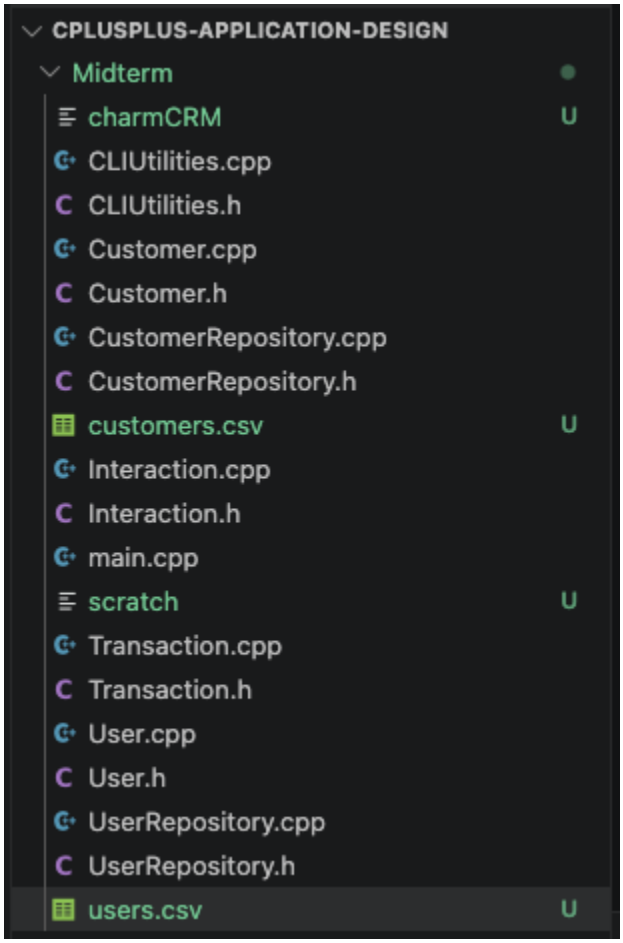
After choosing not to modify the names of either save file (users.csv or customers.csv) - Note: Admin users have the option of changing the name of the .csv output and read by the program upon save. This behavior only functions while the program is in use. After exiting Charm completely the program defaults to looking for "users.csv" and "customers.csv." This will be fixed in future updates - the program notifies us the data was saved successfully and takes users back to the log in menu:

```
 ;) CHARM CRM - Login (1/3) <3
Enter 1 To Log In
Enter 2 To Exit
Choose Option:
```

From here, users can enter "2" to close Charm completely:

Returning to Charm & Loading Data

Notice the files “users.csv” and “customers.csv” now exist in our root directory:



The program checks for the existence and successful opening of “users.csv” and “customers.csv,” and the flag that determines whether initialization occurs is set to false:



During our last use, the flag was successfully flipped after initialization, and because these files exist and open, Charm is able to pick up where it left off. When users open Charm next initialization does not occur and they are taken directly to the log in menu:

```
mayma@Mays-Malik Midterm % ./charmCRM
 User Data Loaded Successfully
 Customer Data Loaded Successfully

  ;) CHARM CRM - Login (1/3) <3
 Enter 1 To Log In
 Enter 2 To Exit
 Choose Option:
```

Notice that user and customer data are found and loaded successfully.
If I log in as Charm's newest Admin employee Lily…

```
 ;) CHARM CRM - Login (1/3) <3
Enter 1 To Log In
Enter 2 To Exit
Choose Option: 1

 ;) CHARM CRM - Login (2/3) <3
Enter Email: lily@charmcrm.com

 ;) CHARM CRM - Login (3/3) <3
Enter Password: 12345
```

I'm taken to the Admin menu, as expected.

```
  ;) CHARM CRM <3
 1. Create New Customer Record
 2. Create New User Record
 3. View Customer Records
 4. View User Records
 5. Find Customer By ID
 6. Find User By ID
 7. Create New Transaction
 8. Create New Interaction
 9. Save And Log Out
 Choose an option:
```

And that's it! Thanks for checking out Charm CRM!

The Future of Charm

Close Range Goals

- Clean up code styling
    - Make sure all braces open on the subsequent line
    - snake_case for bools and counters and camelCase for all other variables
    - Make sure all inputs are handled/sanitized the same way (where appropriate)
    - etc.
- Persisting interactions and transactions
- Persisting file name changes/lookup
- Make interactions and transactions viewable
- Make records editable
- Input validation
- Real password hashing
- Implement additional search functionality
- Update menus for style, voice, and clarity
- Indicate which user is logged in

Mid Range Goals

- Explore agentic development with Claude Code/Chat GPT Codex
- CLI -> GUI
- Migrate from .csv to database

Long Term Goals

- Interactions send automated follow-up reminders to users via email a/o SMS…
- … with sentiment analysis / RAG generated customer summary e.g. "You met Ken Lithgoe at the Stark Gallery opening last weekend. You chatted about custom saltwater fish tanks and told him you know a guy. Follow up?"
- Chat with Charm about customers e.g. "How many gallons was the fish tank Ken Lithgoe mentioned last weekend?" or "Is Ken Lithgoe's first name Kenneth or Kendall?" (assuming you wrote that down in an interaction)
- Email/SMS customers from within Charm directly